Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** MILU* | ***Context:** Bangladeshi Bengali University Student — Story Generation Thesis Deployment*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU evaluates models using accuracy on closed MCQ items via log-likelihood (non-API) or structured JSON parsing (API) [Q48, Q49, Q51, Q52, Q53]. The deployment requires evaluating open-ended Bengali story generation, which demands free-form text output assessed for narrative coherence, register appropriateness, cultural authenticity, and grounding in Bangladeshi named entities. No MILU mechanism produces a score that transfers to story generation evaluation [DATASET-D5 concern]. The authors themselves note log-likelihood may diverge from generation-based and CoT methods [Q85]. The output-form gap is structural: MCQ accuracy measures forced-choice factual recall, not generative cultural grounding. Even if MILU scores were perfectly calibrated to Bangladeshi knowledge (which they are not), they would still not validate story generation quality.

ID	Evidence
Q48	“For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.” (p.5)
Q49	“We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020).” (p.5)
Q51	“"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\” (p.5)
Q52	“The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities.” (p.5)
Q53	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)
D5	৬৫তম ফলি্মফয়োর অ্যাওয়ার্ডস, ২০২০-এ সরো সনিমোটে‌াগ্রাফরি জন্য পুরস্কারটি কে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry

Strengths

- MILU's evaluation infrastructure is well-documented and reproducible across 0/1/5-shot conditions, log-likelihood, and generative-JSON modes [Q46, Q48, Q53], spanning 42–45 models [Q44, Q75]. This breadth means baseline factual-recall comparisons across models are feasible as a screening step.
- Both base and instruct variants are evaluated [Q45], and per-subject/per-language tables are provided [Q99, Q128], enabling fine-grained baseline comparisons within MILU's intended construct.

ID	Evidence
Q44	“We evaluate 42 different models on MILU, including large proprietary models, open-source multilingual models, and popular fine-tuned models specific to Indic languages.” (p.5)
Q45	“Both the base versions and instruction fine-tuned variants of these models, wherever applicable, are evaluated to measure the improvements gained from fine-tuning.” (p.5)
Q46	“All models, except for proprietary models and LLAMA-3.1-405B, are tested under 0-shot, 1-shot, and 5-shot setups.” (p.5)
Q48	“For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.” (p.5)
Q53	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)

Q75	"We evaluate 45 different LLMs and find that the majority of LLMs struggle on MILU, with GPT4o achieving the highest average accuracy." (p.8)
Q99	"Table 9: Detailed subject level statistics of MILU across different languages." (p.20)
Q128	"Table 47: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-9B-IT on MILU across different languages." (p.52)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: No — MILU produces single discrete labels (option1–4) or structured JSON [Q51, Q53]; the deployment requires open-ended free-form Bengali narrative text. Output modalities do not match.

OF-2: Not applicable — deployment is text-only; no TTS requirement. INSUFFICIENT DOCUMENTATION on whether MILU's evaluation infrastructure could be extended to speech, but this is not a deployment requirement.

OF-3: Target population is highly literate (university students [WEB-3]); literacy is not a barrier for either MILU's or the deployment's text format.

OF-4: Severe external-validity violation: MCQ accuracy on India-centric content does not predict Bangladeshi story-generation quality. The deployment must construct a separate generative evaluation rubric (per BanglaSocialBench precedent [WEB-16]).

ID	Evidence
Q51	""For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\" (p.5)
Q53	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
WEB-3	University student literacy supports text-based deployment (https://en.wikipedia.org/wiki/Education_in_Bangladesh)
WEB-16	BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric (https://arxiv.org/abs/2603.15949)

Information Gaps

- No story-generation evaluation methodology exists in MILU; supplementation is structural, not incremental

Requires Expert Verification

- Design and validation of a Bangladeshi story-generation rubric by native-speaker linguists and literary scholars

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Remediation

Gap 1: MCQ accuracy via log-likelihood [Q51] does not transfer to evaluating open-ended Bengali narrative generation

Recommendation: Adopt a generation-based evaluation protocol (per the limitation noted in [Q85]) with multi-criteria human scoring (cultural authenticity, register, named-entity accuracy, narrative coherence) plus optional LLM-as-judge calibration validated against Bangladeshi annotator scores. Treat MILU accuracy purely as an upstream screening signal.

ID	Evidence
Q51	“"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\” (p.5)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)